

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech (EE) Examination****Apr-May 2018****EE245 Control Systems****Date: 02.05.2018, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1** Do as directed. **[08]****(a)** Choose the right answer from the given options.(i) The output signal of any controller is _____. **[01]**

- | | |
|----------------------|---------------------|
| a) Error signal. | b) Feedback Signal. |
| c) Actuating Signal. | d) Output Signal. |

(ii) _____ system is less sensitive to noise present in the system. **[01]**

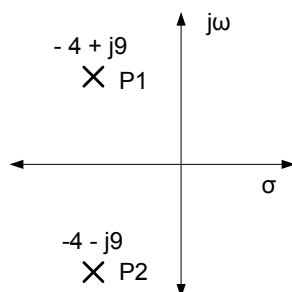
- | | |
|------------------------------------|---------------------------------|
| a) Linear System. | b) Non- Linear System. |
| c) Continuous Time control system. | d) Discrete Time control system |

(iii) Find the appropriate choice for Closed loop control system characteristics. **[01]**

- Accurate , Slow , Sensor
- Accurate, High Bandwidth, Simple in construction.
- Accurate , Difficult for maintenance , Error Detector
- Accurate , Cheap , Feedback Element

(iv) The terms in the first column of the Routh array of the characteristic equation of certain system are 2, 1, 8, -7, -2, 6. The numbers of roots of the characteristic equation in the right half of s-plane are _____. **[01]**

- | | |
|-------|---------------------|
| a) 2. | b) 3. |
| c) 6. | d) None of a),b),c) |

(v) The poles of the system is given in below **Fig.1**, system damped frequency of oscillation is _____ rad/sec. **[01]**

- | | |
|-------|---------------------|
| a) 3. | b) 0.707 |
| c) 5. | d) None of a),b),c) |

Fig.1 Q-1(a-v))(vi) The electrical representation of an Error detector is _____. **[01]**

- | | |
|--------------------------------|-------------------------------|
| a) Op Amp as an Amplifier | b) Op Amp as a Sub tractor |
| c) Op Amp as a pulse generator | d) Op Amp as a differentiator |

(b) Define the following terms. (Any Two). **[02]**

1. Dominant Poles
2. Insignificant Poles
3. Damping
4. Settling Time

- Q - 2** Derive the transfer function of Armature controlled separately excited DC motor and [13]
Field controlled separately excited DC motor with necessary assumptions and derivations.

OR

- Q -2(a)** Find out transfer function of the system whose block diagram is shown in **Fig. 2** using [06]
Block Diagram Reduction method.

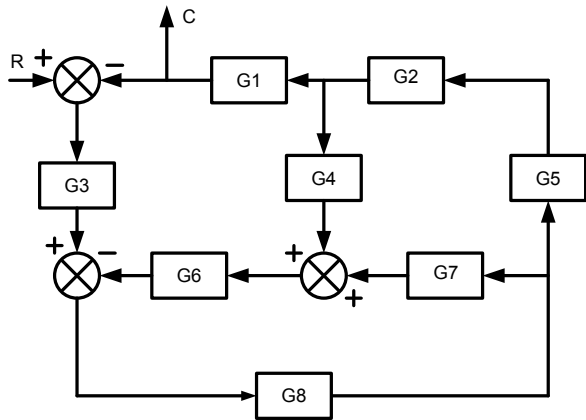


Fig.2 (OR Q-2(a))

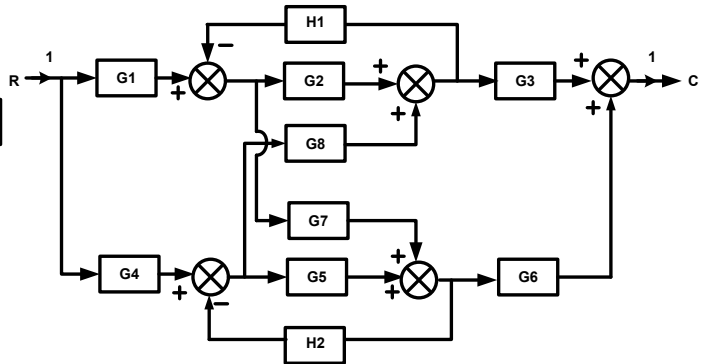


Fig.3 (OR Q-2(b))

- Q -2(b)** From the block diagram shown in **Fig.2**, find out $\frac{C(s)}{R(s)}$ by Mason's Gain Formula. [07]

- Q -3(a)** (i) Find out the angle of departure from the given data from upper complex pole. [03]
Poles : 0 , -2 , -3+2j , -3-2j Zeros : -----

- Q -3(b)** Sketch the root locus of the system on the graph paper with open loop transfer [05]
function of the system is given as $G(s)H(s) = \frac{K}{s(s+1)(s+2)(s+4)}$.

Where K is varying from zero to infinity.

- Q -3(c)** Find the state transition matrix e^{At} for the system whose A matrix is given as [06]
$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$$

OR

- Q -3(a)** A system is described by [07]

$$\dot{x} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u(t) \quad \& \quad y = [1 \quad 2 \quad 0] x(t)$$

Check Controllability and Observability.

- Q -3(b)** Discuss the stability of the system whose characteristic equation is given as [07]
 $s^6 + 3s^5 + 5s^4 + 9s^3 + 8s^2 + 6s + 4 = 0$

SECTION – II**Q - 4** Do as directed. **[08]****(a)** Choose the right answer.**(i)** For the state space representation choose the most appropriate combination. **[01]**

- | | |
|--------------------|----------------------------|
| 1. A matrix | A. Input Matrix |
| 2. B matrix | B. System Matrix |
| 3. C matrix | C. $C [sI - A]^{-1} B + D$ |
| 4. D matrix | D. Output Matrix |
| 5. Transfer Matrix | E. Transmission Matrix |
| a) 1B,2D,3A,4E,5C | b) 1B,2A,3D,4E,5C |
| c) 1D,2B,3E,4A,5C | d) 1B,2D,3E,4A,5C |

(ii) If the system is having “P” numbers of poles and “Z” numbers of Zeros, for the magnitude plot the contribution made by Zeros at origin is given by _____ and for phase angle plot the contribution made by Poles at origin is given by _____.

- | | |
|--|--|
| a) $20 \log \omega, -90^\circ$ | b) $20 \cdot P \log \omega, -90^\circ \cdot P$ |
| c) $20 \cdot P \log \omega, -90^\circ \cdot Z$ | d) None of a), b), c) |

(iii) State true or false. **[01]**

“According to the Bode plot for the stable system either G.M. or P.M. must be positive.”

(iv) If we increase the value of K_p in Proportional controller, the oscillations will _____.

- | | |
|-----------------------|------------------------|
| a) Decreases. | b) Increases. |
| c) Remains unchanged. | d) None of (a),(b),(c) |

(v) System with $\xi = 0$ are called as _____ system. **[01]****(b)** Define the term “Compensator”. Draw the block diagrams of each compensator configurations. **[03]****Q-5(a)** Derive the expression of $C(t)$ for 2nd order under damped system by giving unit step input. **[07]****Q-5(b)** For a control system shown in **Fig 3**, find values of K_1, K_2 so that $M_p = 25\%$ and $T_p = 4$ sec. Assume unit step unit. **[06]**

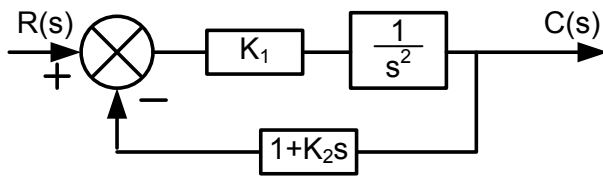


Fig.3 (Q-5(b))

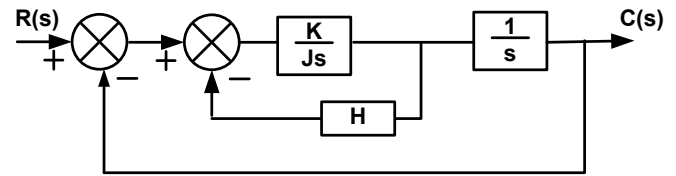


Fig.4 (OR - Q-5(b))

OR

- Q-5(b)** Determine the value of “K” and “H” of the closed loop control system so that the maximum overshoot in unit step response is 25% and the peak time is 2 sec. Assume $J = 1 \text{ kg.m}^2$. Refer **Fig.4** [06]

- Q-6(a)** Sketch the nature of Bode plot on semi log graph paper for the unity feedback control system having whose $G(s)$ is given below, Find out [07]
- (a) Gain cross over frequency (b) Phase cross over frequency
(c) G.M. (d) P.M.
Also Comment on stability.

$$G(s) = \frac{80}{s(s+2)(s+20)}.$$

- Q-6(b)** Sketch the polar plot of the type two system open loop transfer functions is given as [07]

$$G(s) = \frac{K}{s^2(1+sT_1)(1+sT_2)}.$$

OR

- Q-6(a)** The frequency response of a linear system $G(j\omega)$ is provided in the tabular form below. [03]

$ G(j\omega) $	1.3	1.2	1.0	0.8	0.5	0.3
$\angle G(j\omega)$	-130°	-140°	-150°	-160°	180°	-200°

The gain margin and phase margin of the system are _____ and _____.

- a) - 6 dB and 30° b) 6 dB and 30°
c) 6 dB and -30° d) - 6 dB and -30°
- Q-6(b)** Sketch the polar plot of the type two system open loop transfer functions is given as [06]

$$G(s) = \frac{1}{(1+s)(1+2s)}.$$

- Q-6(c)** [05]

Consider a lag compensator $G_c(s)$ defined by $G_c(s) = \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)}$, show that

at a frequency ω_1 , where $\omega_1 = \frac{1}{\sqrt{T_1 T_2}}$ the phase angle of $G_c(j\omega)$ becomes zero.
